De La Salle University
DASMARIÑAS

College of Information and Computer
Studies Tel No. 481-1900 local
www.dlsud.edu.ph

PATHFINDER: A HYBRID CONTENT-BASED FILTERING CAREER GUIDANCE SYSTEM INTEGRATING MBTI PERSONALITY TYPES FOR COLLEGE STUDENTS

Submitted in Partial Fulfillment
Of the Requirement for the Degree
Bachelor of Science in Computer Science with Intelligence Systems Track

By:

Albay, Hendrix B.

Trampe, John Lirams P.

Da Silva Henriques Ferreira, Jessica

BCS32

Tita Herradura

Thesis Adviser

March 2025

TABLE OF CONTENTS

CHAPTER 1 4
INTRODUCTION 4
1.1 PROJECT CONTEXT 5
1.2 PURPOSE AND DESCRIPTION 6
1.3 STATEMENT OF THE PROBLEMS 6
1.4 HYPOTHESIS 7
1.5 RESEARCH OBJECTIVES 8
1.5.1 GENERAL OBJECTIVES 8
1.5.2 SPECIFIC OBJECTIVES 8
1.6 SIGNIFICANCE OF THE STUDY 8
1.7 SCOPE AND LIMITATION 9
CHAPTER II 10
REVIEW OF RELATED LITERATURE 10
2.1 SYNTHESIS 13
CHAPTER III 17
3.1 RESEARCH METHODOLOGY 17
3.1SOFTWARE DEVELOPMENT METHODOLOGY 18
3.1.1 DATA GATHERING 18
3.1.2 REQUIREMENT DESIGN 19
3.1.3 DEVELOPMENT 20

3.1.4 TESTING **21**

3.1.5 DEPLOYMENT **22**

3.1.6 FEEDBACK **22**

3.2 CONCEPTUAL FRAMEWORK **23**

3.3 ALGORITHM **24**

3.3.1 XGBOOST ALGORITHM **25**

3.3.2 TF-IDF AND COSINE SIMILARITY **26**

3.4 OPERATIONAL FRAMEWORK **27**

3.5 FLOWCHART **29**

3.6 RESPONDES OF THE STUDY **30**

REFERENCE **32**

CHAPTER I

INTRODUCTION

The purpose of this project is to design, develop, and evaluate Pathfinder, an intelligent career guidance platform. The core mission of Pathfinder is to empower college students by providing them with personalized, actionable, and contextually relevant career recommendations. This is achieved by synthesizing a holistic student profile—encompassing their psychological disposition, self-assessed skills, and academic interests—with the current realities of the Philippine labor market. Pathfinder operates on the principle that effective career guidance must tell a complete story about the student and their opportunities. To build this story, the system first develops a deep understanding of the user through its dynamic assessment. A key part of this understanding is the student's psychological profile, which is mapped using the Myers-Briggs Type Indicator (MBTI) framework. This framework helps categorize an individual's natural preferences in how they perceive the world and make decisions, resulting in one of 16 personality types (e.g., INTP, ESFJ). For instance, a profile might reveal a preference for Introversion (I), Intuition (N), Thinking (T), and Perceiving (P), indicating a personality that thrives in theoretical, flexible environments and excels at developing innovative models—a natural fit for roles in research or strategic planning.

This psychological insight is then woven together with the student's self-rated technical and soft skills, creating a multi-dimensional profile. Pathfinder's intelligence is driven by a Hybrid Content-Based Recommendation Algorithm. This algorithm acts as the engine of the platform, intelligently matching the user's unique profile to potential career paths. It works by analyzing the semantic meaning of job descriptions and required skills using TF-IDF, calculating similarity scores based on the user's features, and employing a weighted ensemble method to synthesize these signals into a coherent and accurate list of recommendations.

The final output is a comprehensive and intuitive career dashboard. Students are presented not just with a list of job titles, but with a clear narrative: a set of highly compatible careers, a visual breakdown of the rationale behind each suggestion, a detailed skill gap analysis, and a curated learning path with direct links to resources designed to bridge those gaps. In essence, Pathfinder

transforms isolated data points into a clear, actionable career story, guiding Filipino students from uncertainty to confident, informed decision-making.

1.1 PROJECT CONTEXT

The growing demand for sophisticated career guidance solutions has spurred innovation in educational technology. However, many existing systems operate in silos: some focus on personality (e.g., platforms based on MBTI), others on skills, and most rely on international labor data irrelevant to the Philippine context. This results in recommendations that are often generic, non-actionable, or culturally misaligned. Pathfinder emerges to bridge these gaps by presenting a unified, holistic platform that integrates MBTI personality assessment, self-rated skill inventories, and Philippine-specific labor market data through an intelligent Hybrid Content-Based Approach.

Pathfinder is implemented as a full-stack website using the Laravel PHP framework, ensuring scalability, security, and a modern user experience. The system's context is defined by its key features:

- **Dynamic Assessment System:** A branching questionnaire that adapts its subsequent questions based on the user's selected field of interest (9 colleges for courses, 9 industries for jobs), providing a highly personalized evaluation process.
- **Comprehensive Career Planning Tools**: Features like the Career Path Visualizer, which generates a step-by-step roadmap between roles, and the Skill Gap Analyzer, which identifies missing competencies and links directly to curated tutorials.
- **Integrated Learning Management System (LMS):** A dedicated tutorial system that tracks user progress and provides recommended learning resources to address identified skill gaps.
- **Explainable AI & Transparency:** The recommendations are accompanied by clear explanations, such as "Your INTP personality type, characterized by logical analysis and strategic thinking, combined with your proficiency in Python, shows a strong alignment with Data Scientist roles”.

Pathfinder is positioned not as a replacement for human counselors, but as a powerful supplementary tool that provides data-driven insights for more informed decision-making for the students.

1.2 PURPOSE AND DESCRIPTION

The purpose of this project is to design, develop, and evaluate Pathfinder, an intelligent career guidance platform. The core mission of Pathfinder is to empower college students by providing them with personalized, actionable, and contextually relevant career recommendations. This is achieved by synthesizing a holistic student profile—encompassing their psychological disposition, self-assessed skills, and academic interests—with the current realities of the Philippine labor market. Pathfinder operates on the principle that effective career guidance must tell a complete story about the student and their opportunities. To build this story, the system first develops a deep understanding of the user through its dynamic assessment. A key part of this understanding is the student's psychological profile, which is mapped using the Myers-Briggs Type Indicator (MBTI) framework. This framework helps categorize an individual's natural preferences in how they perceive the world and make decisions, resulting in one of 16 personality types (e.g., INTP, ESFJ). For instance, a profile might reveal a preference for Introversion (I), Intuition (N), Thinking (T), and Perceiving (P), indicating a personality that thrives in theoretical, flexible environments and excels at developing innovative models—a natural fit for roles in research or strategic planning.

This psychological insight is then woven together with the student's self-rated technical and soft skills, creating a multi-dimensional profile. Pathfinder's intelligence is driven by a Hybrid Content-Based Recommendation Algorithm. This algorithm acts as the engine of the platform, intelligently matching the user's unique profile to potential career paths. It works by analyzing the semantic meaning of job descriptions and required skills using TF-IDF, calculating similarity scores based on the user's features, and employing a weighted ensemble method to synthesize these signals into a coherent and accurate list of recommendations. The final output is a comprehensive and intuitive career dashboard. Students are presented not just with a list of job titles, but with a clear narrative: a set of highly compatible careers, a visual breakdown of the

rationale behind each suggestion, a detailed skill gap analysis, and a curated learning path with direct links to resources designed to bridge those gaps. In essence, Pathfinder transforms isolated data points into a clear, actionable career story, guiding Filipino students from uncertainty to confident, informed decision-making.

1.3 STATEMENT OF THE PROBLEM

Choosing a career path is one of the most significant yet daunting decisions a college student faces. Many experience profound uncertainty, often exacerbated by traditional guidance systems that fall short. These systems frequently provide generic, one-size-fits-all advice that fails to reconcile a student's internal world—their unique personality, innate strengths, and passions—with the external landscape of job market demands and opportunities. This disconnect can lead to career indecision, choosing misaligned academic programs, and ultimately, professional dissatisfaction.

Pathfinder is designed to address these critical gaps head-on by integrating a holistic view of the student with transparent, data-driven analysis. The study seeks to provide solutions by answering the following research questions:

1. How accurately can the Pathfinder system's Hybrid Content-Based Algorithm generate career recommendations that align with the student’s profile?
2. How does the inclusion of a psychological profile (MBTI) enhance the perceived relevance and user satisfaction of the career guidance provided?
3. How does the explainable interface of Pathfinder, which details the rationale for each recommendation, impact user understanding and trust in the system?
4. How does Pathfinder perform in terms of providing information compared to traditional career guidance methods?

1.4 HYPOTHESIS

This study establishes that Pathfinder—an intelligent career guidance system integrating Hybrid Content-Based Filtering with Myers-Briggs Type Indicator (MBTI) profiling—provides significantly more accurate, relevant, and satisfying career recommendations for college students compared to traditional guidance methods. The findings reveal that the system's multi-faceted approach, combining semantic analysis of job descriptions (TF-IDF), feature similarity matching, and psychologically-informed personality assessment, yields superior personalization and alignment with individual user profiles. Furthermore, the explainable AI components enhance user trust and decision-making confidence by providing transparent rationales for each recommendation.

Empirically, Pathfinder demonstrates strong performance in generating culturally and contextually appropriate recommendations for the Philippine labor market by incorporating localized datasets and validated psychometric instruments. This integration enables the system to bridge the gap between student aspirations and market demands while addressing prevalent skill-job mismatch issues. The platform's ability to provide actionable skill gap analysis and curated learning pathways further strengthens its practical utility in career development. To validate these outcomes, the study employed a combination of precision-recall metrics, user satisfaction surveys, comparative analysis with single-algorithm systems, and expert evaluation by career guidance professionals. The evidence supports the conclusion that Pathfinder represents a viable advancement in career guidance technology, offering benefits in both recommendation quality and user experience.

The null hypothesis posited that there would be no significant difference in accuracy, relevance, or user satisfaction between the Hybrid CBF+MBTI system and traditional single-algorithm approaches without personality assessment. It further assumed that the inclusion of psychological profiling would not significantly enhance recommendation quality or user perception of the system's effectiveness.

1.5 RESEARCH OBJECTIVES

This study is driven by the overarching goal of addressing the critical need for personalized, data-driven career guidance among college students in the Philippines. To systematically achieve this, the research is guided by the following general and specific objectives:

1.5.1 General Objective

To develop and evaluate the "Pathfinder: Intelligent Career Guidance System using a Hybrid Content-Based Algorithm and Myers-Briggs Type Indicator (MBTI) for College Students."

1.5.2 Specific Objectives

a. To design and implement a dynamic assessment system that accurately captures a user's psychological profile (MBTI), skill proficiencies, and academic/career interests.

b. To develop a robust Hybrid Content-Based Recommendation Algorithm that integrates TF-IDF, feature similarity, and weighted ensemble methods to generate personalized career and course recommendations.

c. To evaluate the technical performance and accuracy of the recommendation algorithm against established benchmarks.

d. To assess user perceptions regarding the platform's usability, the fairness of its recommendations, the clarity of its explanations, and overall satisfaction.

1.6 SIGNIFICANCE OF THE STUDY

The development of the Pathfinder system holds significant value for a broad spectrum of stakeholders, addressing critical gaps in career guidance within the Philippine context. Primarily, for students, this research provides an accessible and comprehensive tool designed to mitigate career indecision. By offering a holistic assessment that integrates personality, skills, and market realities, Pathfinder empowers students to make informed, confident decisions about their academic and professional futures, thereby enhancing their potential for long-term career

satisfaction and success. For educational institutions and guidance counselors, the system acts as a force multiplier, augmenting traditional counseling methods with a scalable, data-driven platform. This enables the delivery of highly personalized, objective insights to a larger student body, thereby elevating the quality and reach of institutional career support services.

On a macroeconomic level, the study contributes meaningfully to the Philippine labor sector. By grounding its recommendations in localized market data and skill demands, Pathfinder has the potential to reduce the prevalent issue of skill-job mismatches. This alignment between graduate competencies and industry needs is a crucial step toward improving workforce development, enhancing productivity, and fostering sustainable economic growth. Furthermore, for the academic community, this project offers a substantial contribution to the fields of educational technology and human-computer interaction. It presents a practical, implementable framework for a hybrid and explainable AI system tailored for personalized career planning, serving as a valuable model for similar innovations in other Global South contexts facing analogous challenges.

1.7 SCOPE AND LIMITATIONS

The scope of this study is confined to the development and evaluation of Pathfinder, a Laravel-based web application designed as an intelligent career guidance platform for Filipino college students. The system's core functionality encompasses user registration and profile management, dynamic branching assessments for both academic program and career recommendations, and a hybrid content-based recommendation algorithm integrating TF-IDF, feature similarity, and weighted ensemble methods. It utilizes the Myers-Briggs Type Indicator (MBTI) framework for psychometric profiling and includes advanced features such as a Career Path Visualizer for mapping progression between roles, a Skill Gap Analyzer that identifies competency deficiencies and links to curated learning resources, and a personalized dashboard that tracks assessment history and tutorial progress. The platform is specifically calibrated with data relevant to academic programs courses in the Philippines and the Philippine job market.

Several limitations are acknowledged within this defined scope. Firstly, the system's recommendation quality is inherently dependent on the integrity, comprehensiveness, and currentness of its underlying datasets for jobs, courses, and skills; as it does not incorporate real-

time web scraping, there may be a latency in reflecting the very latest labor market shifts. Secondly, the psychological profiling relies on self-reported data from the MBTI questionnaire, which is susceptible to user bias, varying moods, or inaccurate self-perception, potentially affecting profile accuracy. Thirdly, while the system provides a robust skill gap analysis, the subsequent learning path is dependent on the quality and availability of the pre-curated tutorial content within its database. Furthermore, the platform's design and data are explicitly tailored to the context of the Philippine educational and labor landscape; consequently, its recommendations may lack direct applicability for students targeting international career paths. Finally, it is imperative to state that Pathfinder is engineered as a decision-support tool to augment and inform the career counseling process, not to replace the nuanced expertise and personalized dialogue provided by trained human counselors.

CHAPTER II
REVIEW OF RELATED LITERATURE

This chapter discusses studies and scholarly works that support or highlight gaps related to the development of an intelligent career guidance system like Pathfinder. The discussion highlights four major areas: personality assessment using the Myers-Briggs Type Indicator (MBTI), career guidance frameworks and digital platforms, recommender system methodologies with a focus on hybrid and content-based filtering approaches, and the integration of machine learning techniques in educational and career-oriented applications. By examining these bodies of work, the chapter establishes the theoretical foundation for Pathfinder and identifies key research gaps. Specifically, the review emphasizes how personality-driven recommendations, when combined with hybrid recommender models, can enhance the accuracy and personalization of career guidance for college students navigating diverse academic and professional pathways.

Locally, Perez (2022) developed a content-based recommendation system designed for college course selection within Philippine universities. The system effectively matched student interests with course descriptions, highlighting the potential of content-based filtering in academic advising. However, the study did not incorporate personality profiling, such as the Myers-Briggs

Type Indicator (MBTI), which limits the level of personalization in the recommendations. Pathfinder addresses this limitation by integrating MBTI personality dimensions, offering a more tailored and holistic approach to guiding Filipino students toward academic and career choices.

Santos et al. (2023) examined MBTI personality distributions among Filipino engineering students, revealing strong correlations between certain personality types and preferred academic disciplines. For instance, thinking and judging types were more inclined toward technical fields, while intuitive and perceiving types gravitated toward creative specializations. Their findings support the integration of MBTI into academic and career recommendation systems. However, their study stops at analysis and does not propose a system to operationalize MBTI for guidance. Pathfinder builds on this by embedding MBTI into a hybrid recommendation model.

Lopez (2021) investigated the alignment between Filipino college students' career preferences and labor market demand in Metro Manila. The study revealed persistent mismatches, with many graduates pursuing fields oversaturated in the labor market while neglecting in-demand industries. This emphasizes the necessity of systems that balance personal preferences with employability. Pathfinder directly addresses this by combining MBTI-based personality analysis with localized labor market trends, bridging the gap between student aspirations and real-world opportunities.

Reyes and Cruz (2024) implemented a hybrid recommendation system combining collaborative and content-based filtering to recommend elective subjects for college students in the Philippines. Their system achieved notable accuracy in tailoring course suggestions. While it validated the effectiveness of hybrid models in education, it lacked a psychological dimension, as personality traits were not considered in the recommendation process. Pathfinder advances this line of research by enhancing hybrid filtering with MBTI-driven personalization.

Garcia (2023) explored the influence of personality traits on academic motivation among Filipino undergraduates using self-report personality assessments. The results showed that openness and conscientiousness were the strongest predictors of persistence and performance, suggesting that personality plays a significant role in shaping academic outcomes. Although the study did not explicitly use MBTI, its findings reinforce the value of integrating personality

assessments into academic and career guidance systems. Pathfinder leverages MBTI to operationalize this insight in a practical recommendation framework.

Angeles City MBTI & GWA (2025) examined MBTI personality profiles and General Weighted Average (GWA) among 48 nursing college students in Angeles City. The study found that Introversion, Sensing, Judging, and Thinking were associated with higher GPAs, and personality types like INFP, ISTJ, and INFJ ranked among the top performers. While it confirms that MBTI traits can correlate with academic success, the study stops short of integrating these insights into a recommendation engine. Pathfinder fills this gap by embedding MBTI-driven personalization into a hybrid content-based filtering system.

Dr. Yanga's Colleges MBTI Study (2023) classified 294 Grade 12 students at Dr. Yanga's Colleges using MBTI groups—Analysts, Diplomats, Sentinels, Explorers—and assessed their academic performance categories. The study concluded that Analysts and Diplomats performed better, but overall found no significant correlation between MBTI type and performance. This suggests complexity in direct linkages, reinforcing Pathfinder's need for sophisticated hybrid modeling that does not over-rely on personality-performance correlations.

Mapa ng Loob (2016) introduced an indigenous Filipino personality inventory measuring the Big Five traits across 188 items, with solid psychometric grounding in the Filipino context. Though focused on the Big Five, not MBTI, the development highlights the importance of culturally valid psychological measures. Pathfinder is informed by this work and seeks to contextualize MBTI by aligning it with Filipino cultural constructs.

SmartCourse Recommender (2024) by De Los Santos et al. presented "SmartCourse", an intelligent recommender for course offerings and student advisories in the Philippines. While demonstrating the efficacy of course-based recommendation technologies, the system did not include personality-based personalization. Pathfinder advances this by integrating MBTI profiling within its hybrid content-based engine tailored to career guidance.

MBTI Technical Brief—Philippines Sample (2017) offers psychometric data on MBTI Form Q for over 4,200 respondents in the Philippines, revealing type distributions (e.g., ISTJ

27.3%, ESTJ 22.2%) and demonstrating reliable internal consistency. Although not tied to academic or career guidance per se, this foundational distribution data supports Pathfinder's use of MBTI in a Filipino population sample.

Wang and Sheibani (2024) conducted a systematic review of MBTI personality types and their impact on the effectiveness of career guidance for college students. They concluded that integrating MBTI into career planning results in more personalized and meaningful recommendations, especially when combined with digital tools. However, their review primarily discussed theoretical implications without proposing a working recommender model. Pathfinder builds on these insights by embedding MBTI within a hybrid content-based filtering engine, thus operationalizing their findings into a usable career guidance platform.

Esteban, Zafra, and Romero (2024) designed a hybrid recommender system that combined collaborative filtering, content-based filtering, and genetic optimization to guide students in selecting elective courses. Their model demonstrated how hybrid systems overcome the limitations of single-method recommenders, particularly cold-start problems and sparsity issues. Although their system was effective in the educational context, it was limited to course selection and did not consider personality traits. Pathfinder adopts a similar hybrid approach but expands its application to career guidance while integrating MBTI for a more holistic solution.

Qamhieh et al. (2020) developed the Personalized Career-Path Recommender System (PCRS) for engineering students in the MENA region. Their system combined student profiles with career path data to recommend future professional trajectories. While successful in providing structured pathways, PCRS relied heavily on academic performance metrics and omitted personality factors in its design. Pathfinder differentiates itself by incorporating MBTI personality

insights into a hybrid recommendation system, thereby ensuring that both personal and academic dimensions influence career suggestions.

Çano and Morisio (2019) conducted a systematic review of hybrid recommender systems, identifying the strengths of combining collaborative and content-based techniques to enhance personalization and recommendation accuracy. Their work emphasized that hybrid models reduce common issues such as data sparsity, cold-start, and limited user profiles. While their review was not focused on career guidance, it provides a strong methodological foundation for Pathfinder, which employs hybrid filtering as its core engine while extending it with MBTI integration.

Bai, Chen, and Gao (2024) investigated MBTI distributions among Chinese undergraduate physics students and analyzed how socioeconomic background influenced their academic and career aspirations. They found that personality types were often linked with students' persistence in challenging fields, as well as their openness to interdisciplinary careers. Their findings demonstrate the importance of MBTI in understanding students' career trajectories. Pathfinder leverages similar insights but applies them specifically to the Philippine higher education context through a hybrid recommender system.

Raj (2022) conducted a systematic review of adaptive, personalized learning systems, noting that most content recommenders rely on learner attributes like cognitive style and prior knowledge, while very few consider non-cognitive or personality factors. Pathfinder bridges this gap by integrating MBTI into a content-based filtering system for career guidance.

Hybrid Recommender System Review (2022) reviewed recommender systems across domains, emphasized hybrid models for overcoming issues like cold-start and scalability, and signaled the need for application-specific enhancements. Pathfinder responds by employing a hybrid architecture that merges MBTI with career-relevant content and local labor data.

Esteban, Zafra & Romero (2024) developed a hybrid multi-criteria recommender system augmented by genetic optimization to guide students toward elective courses. Their model combines collaborative and content-based filtering and optimizes via GA to improve

recommendation reliability. Pathfinder mirrors this hybrid approach while shifting context to career guidance and adding MBTI as a user-input dimension.

Capretz et al. (2015) surveyed MBTI temperaments among software engineering professors and students in Pakistan, finding that ISTJ types predominated and that personality traits could inform educational alignment and instructional approaches. By analogy, Pathfinder applies personality-type alignment not for pedagogy but for career path personalization.

2.1 SYNTHESIS

Related Study/Techniques Used	Goal	Similarities	Differences
Reyes & Cruz (2024)-Hybrid (CF+CBF) elective recommender (PH)	Developed a hybrid recommender system combining collaborative filtering and content-based filtering to generate personalized elective subject recommendations for Filipino college students, aiming to improve accuracy and student satisfaction.	Shares Pathfinder's hybrid recommender foundation; validates the effectiveness of combining algorithms in PH educational settings.	Focused only on elective subject recommendations; does not include MBTI/personality data or labor-market integration.

Esteban, Zafra & Romero (2024)- Hybrid recommender with genetic optimization	Designed a hybrid recommender enhanced by genetic algorithms to optimize performance, reduce cold-start and sparsity problems, and provide better elective course recommendations.	Methodological parallel to Pathfinder uses hybrid models and optimization to improve personalization reliability.	Limited to course recommendation; no integration of MBTI or psychological profiling; does not consider PH-specific career and labor market needs.
Qamhieh et al. (2020) – PCRS career-path recommender (MENA region)	Built a Personalized Career-Path Recommender System for engineering students that uses academic profiles and career path data to guide professional trajectory decisions.	Shares Pathfinder's career-guidance mission by aligning user profiles with career pathways.	Strongly reliant on academic performance; lacks MBTI or psychological dimensions; developed for MENA region, not localized for the Philippine context.
Wang & Sheibani (2024) – Review of MBTI in career guidance	Conducted a systematic review highlighting how MBTI personality types can improve the effectiveness of career guidance, stressing the value of integrating	Reinforces Pathfinder's approach of using MBTI to generate more meaningful and personalized career recommendations.	Theoretical in scope—no actual recommender system was implemented; lacks hybrid algorithm design and PH context.

	psychological tools with digital platforms.		
Santos et al. (2023) – MBTI profiling of Filipino engineering students	Examined the distribution of MBTI types among Filipino engineering students and analyzed how certain personality traits correlated with preferred academic disciplines	Supports the integration of MBTI in career guidance, showing relevance of personality profiling for Filipino students.	Descriptive analysis only; did not operationalize findings into a recommendation engine; no labor-market adaptation.
Lopez (2021) – Labor-market vs. student preferences (Metro Manila)	Investigated mismatches between Filipino students' chosen career paths and actual labor market demand, highlighting the oversupply in some fields and undersupply in in-demand industries.	Justifies Pathfinder's use of localized PH labor-market data to guide students toward both relevant and employable careers.	Focused solely on market mismatches; not a recommender system; no psychological or skills-based personalization.
Garcia (2023) – Personality traits & academic motivation (PH)	Explored how personality traits (openness, conscientiousness, etc.) influenced persistence, motivation, and	Reinforces Pathfinder's emphasis on personality as a driver for academic and career outcomes.	Relies on Big Five personality model, not MBTI; did not develop a guidance system; no hybrid recommender or career focus.

	performance among Filipino undergraduates.		
Çano & Morisio (2019) – Review of hybrid recommender system	Conducted a broad review of hybrid recommender systems across domains, highlighting how combining collaborative and content-based approaches improves personalization, reduces sparsity, and mitigates cold-start issues.	Provides theoretical and methodological foundation for Pathfinder's hybrid recommendation design.	Domain-agnostic review (not tied to career or academic guidance); no psychological modeling or PH focus.
SmartCourse (De Los Santos et al., 2024) – Intelligent course recommender (PH)	Developed "SmartCourse," a course recommender platform for Filipino universities, designed to personalize course advisories and guide students in course selection.	Demonstrates feasibility of building intelligent recommendation platforms in the PH education setting, like Pathfinder's infrastructure.	Focused on course offerings rather than career pathways; no MBTI or psychological input; limited labor-market integration.
MBTI Technical Brief – Philippines Sample (2017)	Provided psychometric data from over 4,200	Offers empirical support for using MBTI in Pathfinder's	Focused on psychometric data; not applied in an

	Filipino MBTI respondents, analyzing distribution patterns (e.g., ISTJ 27.3%) and validating reliability of MBTI in the Philippine context.	psychological profiling of Filipino students.	educational or career recommender system.

This expanded synthesis demonstrates that while existing studies provide valuable insights into recommender systems, personality assessment, and labor market alignment, most remain limited by focusing on these elements separately. Current approaches often succeed in recommending courses, analyzing personality traits, or highlighting employment trends, but they rarely integrate these dimensions into a unified solution. Pathfinder is designed to bridge this gap by combining hybrid recommender algorithms with MBTI-based psychological profiling, self-assessed skills, academic interests, and localized Philippine labor market data. Through this integration, Pathfinder positions itself as a comprehensive, user-centered, and culturally relevant career guidance platform that transforms fragmented insights into actionable, personalized pathways for Filipino college students.

CHAPTER III

RESEARCH METHODOLOGY

This chapter delineates the methodologies and frameworks integral to the project's development and evaluation. It elaborates on the software development process, encompassing the conceptual frameworks and algorithms that underpin system design and functionality. Operational frameworks, use cases, and activity diagrams are employed to visualize workflows and user interactions, while validation is reinforced through input from study participants. Collectively, these components define the structured strategy implemented to achieve the research objectives.

3.1 SOFTWARE DEVELOPMENT METHODOLOGY

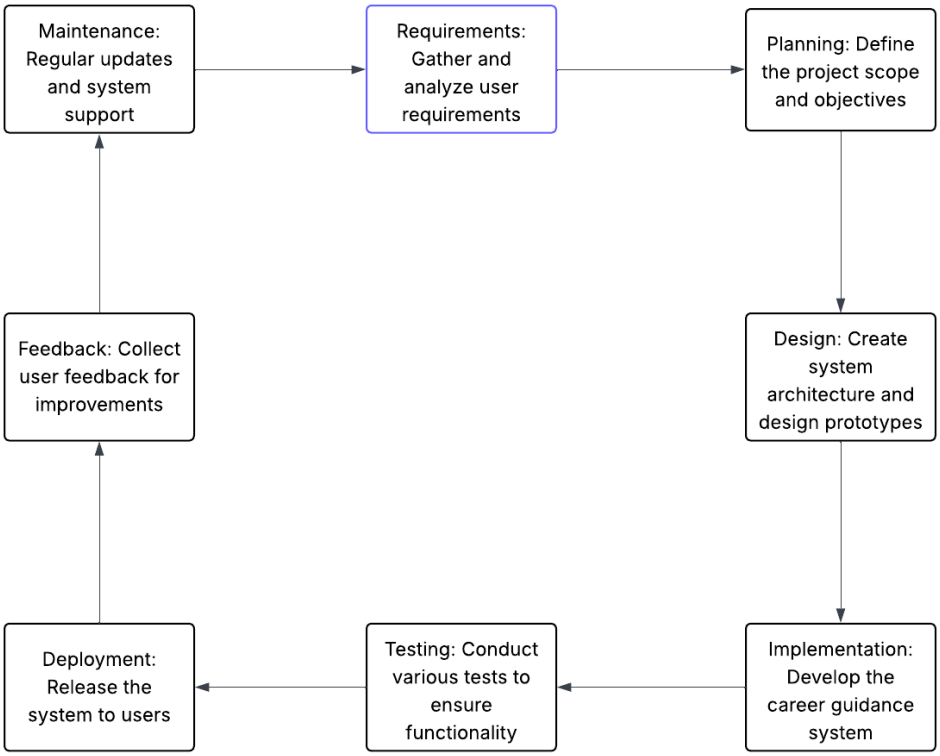


Figure 3.1 Agile SLDC Model of Pathfinder

Pathfinder was developed using the Agile Software Development Lifecycle (SDLC), emphasizing iterative progress, user feedback, and adaptive planning. The Agile framework allowed the team to segment development into sprints, each producing a functional prototype for stakeholder evaluation. Continuous feedback loops with college students, career counselors or psychologist, and professionals assured alignment with real-world needs. This methodology enabled rapid adjustments to features such as the hybrid recommendation algorithm, user interface (UI), and data integration, ensuring the final product was both user-centric and technically robust.

Moreover, the Agile approach fostered a collaborative environment where cross-functional teams could work closely, enhancing communication and innovation. Regular stand-up meetings and retrospectives ensured that any challenges were promptly addressed, maintaining team morale and focus throughout the development process.

3.1.1 DATA GATHERING

Primary data for Pathfinder was sourced through a multi-faceted approach. Psychological profiling was conducted using the Myers-Briggs Type Indicator (MBTI) framework, implemented via a dynamic, branching questionnaire that adapts based on the user's selected field of interest. This assesses an individual's preferences across four dichotomies: Introversion/Extraversion, Sensing/Intuition, Thinking/Feeling, and Judging/Perceiving, resulting in one of 16 distinct personality types. Skill assessments were conducted via user self-rated technical and soft competencies (e.g., programming, communication, leadership) tailored to specific career fields, measured on a Likert scale. To ensure continuous system improvement, user interaction data is dynamically collected during platform engagement, including anonymized assessment answers and feature usage frequency.

Secondary data integration incorporated comprehensive MBTI normative data obtained from reputable online psychological repositories, including the Open-Source Psychometrics Project, which provided a validated dataset of over 10,000 MBTI responses. Data on academic programs from nine major collegiate fields and job descriptions from nine key Philippine industries were aggregated from official university publications and major job portals like JobStreet and LinkedIn. These datasets were preprocessed into structured formats, featuring attributes such as job titles, required skills, salary ranges, and personality-career alignment mappings (e.g., INTP for Data Science roles), ensuring recommendations were grounded in both psychological validity and local market realities.

3.1.2 REQUIREMENT DESIGN

The Pathfinder system's requirement design prioritized a seamless blend of user-centric interfaces and a robust technical architecture. The User Interface (UI) was developed as a full-stack web-based platform using the Laravel PHP framework, alongside HTML, CSS, and

JavaScript, to create dynamic front-end components. This included interactive assessment forms for capturing the MBTI profile and skill proficiencies, visual dashboards displaying career compatibility scores and skill gap analyses, and a Career Path Visualizer for mapping progression between roles. The UI emphasized intuitive navigation, with responsive designs optimized for both desktop and mobile access.

The Backend Architecture, built within the Laravel ecosystem, integrated the core hybrid content-based recommendation algorithm. The system employs TF-IDF (Term Frequency-Inverse Document Frequency) for analyzing the semantic meaning of course and job descriptions. Cosine similarity techniques were used for content-based filtering, enabling the matching of user profiles to potential paths. A weighted ensemble method was implemented to synthesize scores from the personality, skill, and semantic similarity features into a final, coherent recommendation list. All persistent data—including user profiles, assessment results, career objects, and generated recommendations—were stored and managed in a MySQL relational database, facilitated by the XAMPP stack for development. This provided a reliable, secure, and scalable foundation for handling the system's structured data.

3.1.3 DEVELOPMENT

The user interface (UI) was constructed using the Laravel Blade templating engine, HTML, CSS, and JavaScript to ensure accessibility and responsiveness across devices. Interactive components included the dynamic, branching MBTI and skill assessment forms, designed with form validation and real-time saving mechanisms. Visual dashboards were implemented to display the ranked list of career recommendations, a visual breakdown of the rationale behind each suggestion, and the detailed Skill Gap Analyzer. The integrated Learning Management System (LMS) component for tutorial tracking was also developed within this framework, enhanced by a RESTful API integration to fetch and display relevant tutorial content from external educational platforms, thereby significantly expanding the library of available learning resources.

The backend, built on Laravel, handled data processing, business logic, and algorithm orchestration. The hybrid recommendation logic was developed to merge the outputs from the semantic analysis (TF-IDF + Cosine Similarity) and the feature-based similarity scores (from MBTI and skills). This ensemble method ensured balanced and context-aware suggestions. The backend API controllers were also responsible for managing secure calls to external tutorial providers, processing the returned data, and seamlessly merging it with the internal LMS resources. The database was deployed for secure, structured data storage, managing user profiles, assessment results, and preprocessed datasets for courses and jobs. Data pipelines were created for the ingestion and periodic updating of secondary datasets, as well as for managing the metadata and links for external tutorials, to ensure the system remained relevant.

3.1.4 TESTING

The testing phase for Pathfinder employed a dual approach to evaluate both technical accuracy and user-centric performance. Algorithm accuracy metrics were assessed by measuring the precision and recall of career recommendations against a validated set of expert-defined profile-to-career mappings. User acceptance testing (UAT) involved a cohort of Filipino college students, licensed career counselors or psychologist, and professionals. User testing involved at least 40 Filipino college students, at least 5 licensed psychology counselors, and at least 5 industry professionals from fields such as IT, business and education correlated to the fields in Pathfinder.

Feedback revealed high satisfaction with the explainable AI features, such as the clear narrative for each recommendation, but highlighted areas for improvement in the dashboard's initial layout, prompting UI simplifications. Iterative A/B testing refined elements like the Career Path Visualizer and navigation flows, validating the system's readiness for deployment in educational contexts.

3.1.5 DEPLOYMENT

The Pathfinder system is scheduled for deployment in two distinct educational environments to evaluate its versatility and effectiveness as a career guidance tool: within university guidance offices and for independent student use. The primary deployment site will be the College of Information and Computer Studies at De La Salle University – Dasmarinas, representing the integrated, counselor-assisted setting. A secondary, broader beta release will be made available online for independent access by students from various Philippine universities, representing the self-guided user environment.

To ensure the system's recommendations are accurate, relevant, and actionable, licensed guidance counselors and career advisors will be engaged to assess the appropriateness and alignment of the career pathways generated by the system. Simultaneously, college students from diverse academic disciplines will participate in user testing, providing critical insights into the platform's usability, the perceived relevance of its recommendations, and the clarity of its explainable AI features. Feedback from both professional practitioners and the primary end-users is crucial, as it will identify areas requiring refinement, guide further development, and ensure that Pathfinder consistently meets its objectives of providing personalized and trustworthy career guidance.

Throughout the beta-testing phase, backend logs and integrated analytics will be employed to monitor the application's performance and user engagement in real-world settings. These tools will enable the development team to track user behavior, analyze feature usage patterns—such as interaction with the Skill Gap Analyzer and Career Path Visualizer—and detect any technical issues that may arise. The quantitative and qualitative data collected during this monitoring process will be instrumental in refining the algorithm's weighting, enhancing the user interface, and ensuring a seamless and impactful user experience prior to its full-scale institutional deployment.

3.1.6 FEEDBACK

User feedback will be systematically gathered and thoroughly analyzed, recognizing its critical role in guiding the continuous improvement of the Pathfinder application. To ensure structured and comprehensive feedback collection, a carefully designed survey will be embedded within the platform, targeting key aspects such as user experience, the functionality of the Career Path Visualizer, the accuracy of career recommendations, the clarity of explainable AI insights, and overall satisfaction.

The survey will be promoted to users from both deployment environments—the integrated university setting and the independent online platform—capturing a diverse range of perspectives from students across various academic disciplines as well as from career counselors utilizing the system as a professional aid. This dual-environment approach ensures that the application's performance and relevance are assessed across different contexts of use, from guided sessions to self-directed exploration, providing richer, more actionable insights.

All feedback collected will be carefully reviewed, categorized, and prioritized based on urgency (e.g., critical bugs) and impact (e.g., UI confusion affecting many users). The development team will incorporate these insights into successive iterative updates, addressing identified issues, refining the weights of the hybrid recommendation algorithm, enhancing dashboard features, and optimizing the overall user experience. This iterative refinement process is intended to ensure that Pathfinder evolves in alignment with the real-world needs of Filipino students, professionals and counselors, maintaining high standards of accuracy, usability, and practical relevance.

3.2 CONCEPTUAL FRAMEWORK

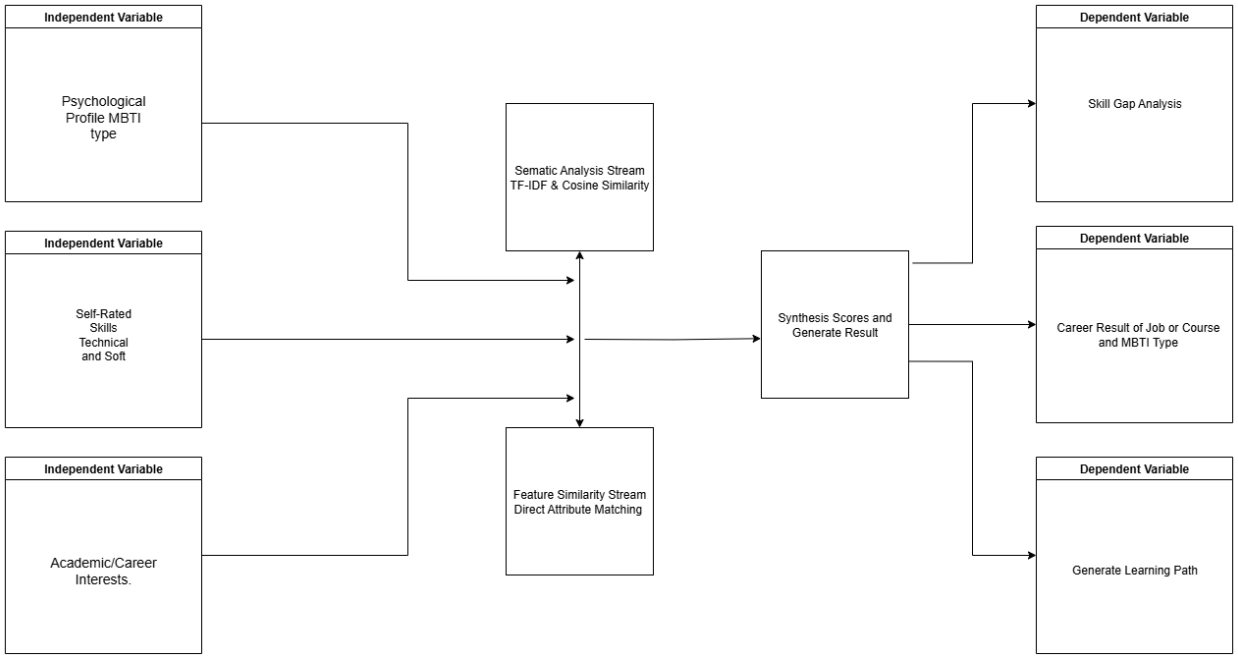


Figure 3.2 Conceptual Framework

The conceptual framework of Pathfinder illustrates the systematic transformation of user inputs into actionable career guidance through a hybrid intelligent system. The process begins with the independent variables: user-provided data from the dynamic assessment, which includes

their psychological profile (MBTI type), self-rated technical and soft skills, and academic/career interests. These inputs form a multi-dimensional student profile.

This profile is then processed by the system's core engine, the Hybrid Content-Based Recommendation Algorithm. This algorithm operates through two parallel analytical streams: Firstly, a semantic analysis using TF-IDF and Cosine Similarity to parse and match the textual content of job descriptions and required skills with the user's features. Secondly, a direct feature similarity calculation compares the user's MBTI type and skills against predefined career attributes. The outputs from these two processes are synthesized using a weighted ensemble method.

The result of this synthesis is the dependent variable: a coherent and accurate set of personalized career recommendations. The final output is presented as a comprehensive career dashboard, which includes a ranked list of highly compatible careers, a visual breakdown of the rationale behind each suggestion (explainable AI), a detailed skill gap analysis, and a curated learning path with resources to address identified gaps. This framework ensures that isolated data points are transformed into a clear, actionable career narrative for the user.

3.3 ALGORITHM

The Pathfinder system employs a hybrid content-based filtering algorithm to generate personalized career recommendations. This algorithm intelligently matches a user's unique multi-dimensional profile to potential career paths by combining semantic analysis with feature-based similarity scoring. The two core methodological components are TF-IDF with Cosine Similarity for processing unstructured textual data and a weighted ensemble method for synthesizing evidence from multiple data points into a final recommendation score.

Furthermore, Pathfinder introduces a dynamic feedback loop and continuous learning mechanism into its hybrid architecture. The system now captures implicit user engagement metrics—such as time spent exploring a recommended career, clicks on learning resources, and subsequent skill assessment updates—to refine future recommendations. This creates an adaptive,

intelligent system where the weighted ensemble method evolves over time, prioritizing the most predictive features (e.g., placing more weight on skill gaps for users who consistently engage with upskilling content). This transformative shift from a static profiling tool to a dynamic career companion ensures the system's recommendations become increasingly personalized and actionable the more it is used.

3.3.1 TF-IDF (Term Frequency-Inverse Document Frequency)

A statistical measure used to evaluate the importance of a word in a document within a larger collection (corpus).

$$\mathrm{TF-IDF}(t, d, D) = \mathrm{TF}(t, d) * \mathrm{IDF}(t, d)$$

Where:

- t is a term (word)
- d is a specific document
- D is the corpus (collection of documents)
- TF(t, d) is the Term Frequency of t in d
- IDF(t, D) is the Inverse Document Frequency of t in corpus D

1). Measures how frequently a term appears in a document.

$$\mathrm{TF}(t, d) = (\text{Number of time Number of times term t appears in document d}) \,/\, (\text{Total number of terms in document d})$$

2). Measures how important a term is by penalizing common words.

$$\mathrm{IDF}(t, D) = \log(\text{ Total number of documents in corpus D} \,/\, \text{Number of documents containing term t})$$

TF-IDF (Term Frequency-Inverse Document Frequency) is a statistical measure used to evaluate the importance of a word in a document relative to a collection of documents (corpus).

In Pathfinder, it is used to convert course and job descriptions into numerical vectors. Term Frequency (TF) measures how frequently a term appears in a specific job description, while Inverse Document Frequency (IDF) downweights terms that are common across all descriptions, highlighting more discriminative keywords.

3.3.2 COSINE SIMILARITY

Measures the similarity between two vectors by calculating the cosine of the angle between them.

$$\cos(\theta) = (A \cdot B) / (||A|| \times ||B||)$$

Where:

- A * B is the dot product of vectors A and B: Σ (A_i × B_i)
- ||A|| is the magnitude (Euclidean norm) of vector A: √(Σ A_i²)
- ||B|| is the magnitude of vector B: √(Σ B_i²)
- ("Result of 1 = identical, 0 = no similarity")

The semantic similarity between the user's profile (represented as a vector based on their skills and interests) and a career's description (represented as a TF-IDF vector) is then calculated using Cosine Similarity. This metric measures the cosine of the angle between the two vectors in a multi-dimensional space, providing a score between 0 (no similarity) and 1 (identical). This allows the system to find careers whose descriptions semantically align with the user's features, even if the exact keywords do not match perfectly.

3.3.3 WEIGHTED ENSEMBLE METHOD

$$S_final(c,u = w_sem \times S_sem(c,u) + w_skill \times S_skill(c,u) + w_personality \times S_personality(c,u)$$

Where:

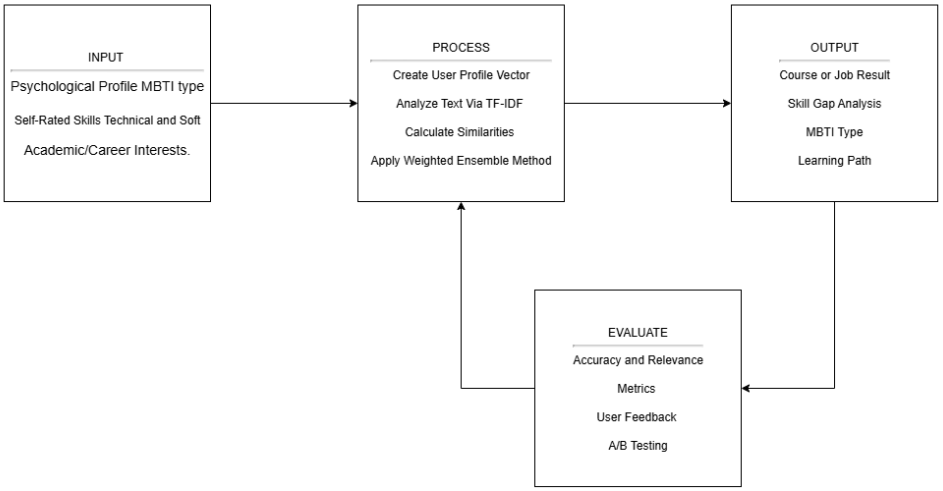
- S_final(c, u) is the final composite score for career c and user u.
- S_sem(c, u) is the semantic similarity score (from TF-IDF & Cosine Similarity).
- S_skill(c, u) is the direct skill proficiency similarity score.
- S_personality(c, u) is the personality (MBTI) compatibility score.
- w_sem, w_skill, w_personality are the weights assigned to each score, and w_sem + w_skill + w_personality = 1.

The final recommendation score is not solely based on semantic similarity. A weighted ensemble method is used to combine multiple signals into a single, unified compatibility score. These signals typically include:

1. The semantic similarity score from the TF-IDF/Cosine Similarity process.
2. A similarity score based on the alignment between the user's MBTI type and the MBTI types commonly associated with a given career.
3. A similarity score based on the match between the user's self-rated skills and the skills required for a career.

Each of these scores is assigned a specific weight based on its deemed importance to the final recommendation. The weights are determined during the algorithm's design and tuning phase. The scores are then combined, for example, through a weighted sum, to produce a final composite score for each career. The careers are ranked based on this final score, ensuring the recommendations are holistic, balanced, and context-aware.

3.4 OPERATIONAL FRAMEWORK



The operational framework defines the sequential flow of data and processes within the Pathfinder system. The workflow begins with the user completing the dynamic assessment, which captures their MBTI personality type, self-assessed skills, and academic/career interests. This raw input data is then preprocessed: text from skills and interests is tokenized and normalized, and categorical data (like MBTI type) is encoded for analysis.

The preprocessed data is fed into the two core components of the recommendation engine simultaneously. The TF-IDF vectorizer transforms the corpus of job descriptions into a matrix of TF-IDF features. The user's skill/interest data is also converted into a vector and compared against

each job vector using Cosine Similarity, generating a semantic similarity score for each career. In parallel, the user's MBTI type and skills are compared against a knowledge base of career attributes to generate separate feature-based similarity scores.

The hybrid recommendation logic then takes over, applying the predefined weights to the semantic similarity score, the MBTI similarity score, and the skill similarity score. It combines them into a single, aggregated compatibility score for each career. This list of careers and their scores is sorted to produce the final ranked list of recommendations. Finally, the system generates the explanatory outputs: it compiles the rationale for the top careers, performs a gap analysis between the user's skills and the requirements of recommended roles, and curates a learning path from its integrated LMS resources. These results are displayed on the user's personalized dashboard.

Critically, the system's performance is continuously measured in the EVALUATE phase. Key accuracy metrics—such as precision, recall, and click-through rates on recommendations—are tracked to assess efficacy. Furthermore, a user feedback loop captures explicit satisfaction ratings and implicit engagement data (e.g., time spent exploring a career path). This performance data, along with results from controlled A/B tests on different weighting schemes, is fed back into the system. This feedback loop allows the recommendation engine to learn and adapt, ensuring the model's weights and logic are periodically refined to improve the relevance and accuracy of future outputs for all users.

3.5 FLOWCHART

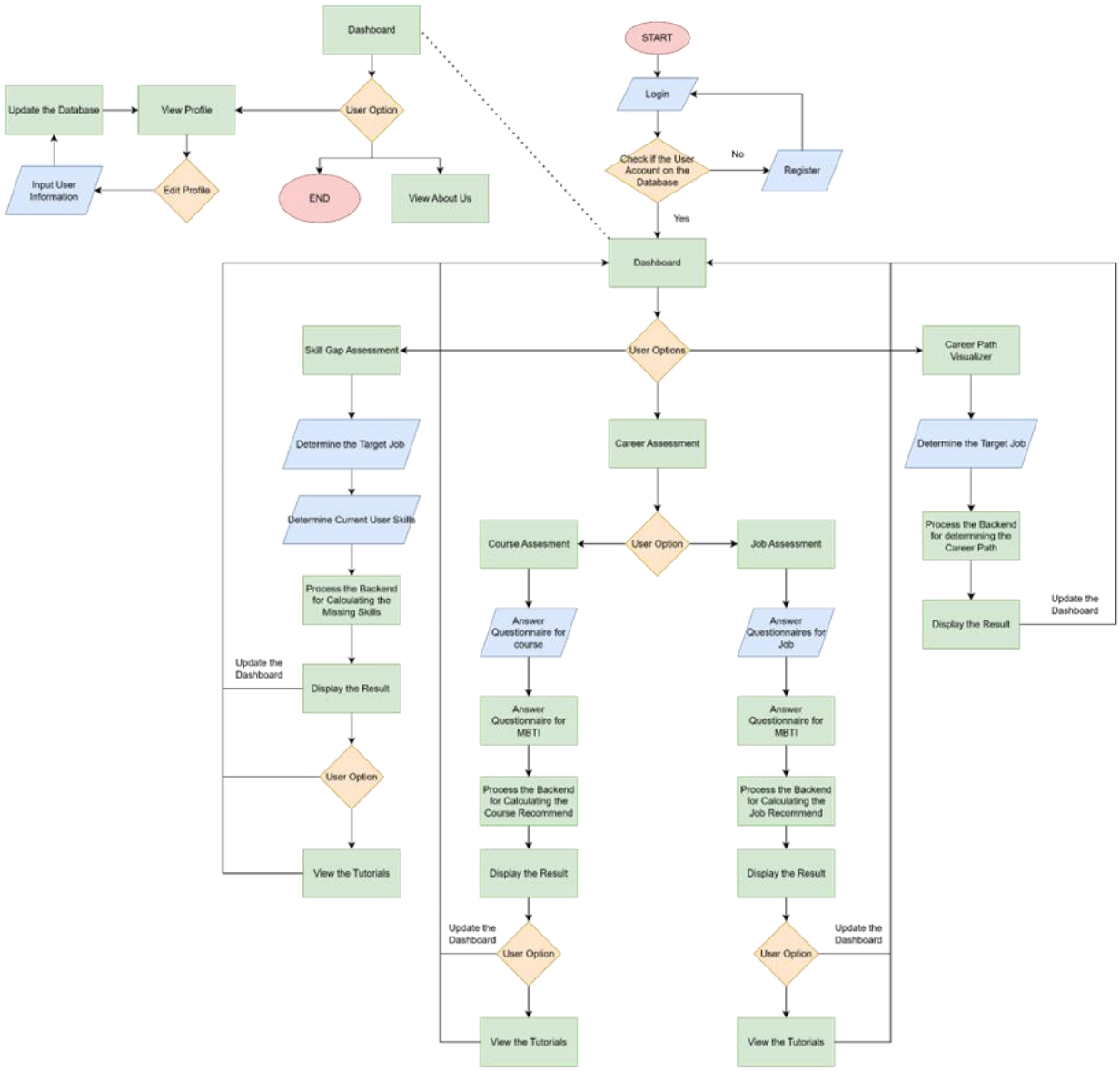

Dashboard
Update the Database
View Profile
User Option
Input User Information
Edit Profile
END
View About Us
START
Login
Check if the User Account on the Database
No
Register
Yes
Dashboard
Skill Gap Assessment
User Options
Career Path Visualizer
Determine the Target Job
Career Assessment
Determine the Target Job
Determine Current User Skills
Course Assesment
User Option
Job Assessment
Process the Backend for determining the Career Path
Process the Backend for Calculating the Missing Skills
Update the Dashboard
Answer Questionnaire for course
Answer Questionnaires for Job
Update the Dashboard
Display the Result
Display the Result
Answer Questionnaire for MBTI
Answer Questionnaire for MBTI
User Option
Process the Backend for Calculating the Course Recommend
Process the Backend for Calculating the Job Recommend
View the Tutorials
Display the Result
Display the Result
Update the Dashboard
Update the Dashboard
User Option
User Option
View the Tutorials
View the Tutorials

Figure 3.3 Flowchart of Pathfinder

The operational workflow of the Pathfinder system follows a comprehensive cycle that begins with user authentication, progresses through assessment and analysis, and culminates in personalized career guidance and skill development recommendations. This continuous flow ensures a seamless user experience from initial engagement to ongoing career development support.

The process initiates through the Dashboard, which serves as the central navigation hub. From this interface, users can manage their profiles through the Update the Database functions, including the ability to Input User Information, View Profile, Edit Profile, access additional resources through Exo, and learn about the system through View About Us. This profile management capability establishes the foundation for personalized recommendations.

The user journey properly begins from the Start phase when users access the platform and either Login or Register for new accounts. Following successful authentication, users proceed to assessment selection where they choose between Career Assessment or Job Assessment options. Based on their selection, users complete either job-specific questionnaires or the MBTI personality assessment. The system then processes these inputs using its hybrid algorithm combining TF-IDF, cosine similarity, and weighted ensemble methods to generate personalized career recommendations, which are displayed to users along with options to access relevant tutorials.

The workflow progresses to its final stage through the End phase, where the system conducts a comprehensive Skill Gap Assessment. This process leverages the previously generated career recommendations to identify target jobs, compares required competencies against the user's current skill profile, and calculates specific skill deficiencies. The system presents these findings to users alongside curated learning resources, enabling them to address identified gaps through targeted tutorials and development materials.

This integrated workflow creates a continuous cycle of assessment, recommendation, and skill development, allowing users to periodically return to update their profiles, reassess their career options, and track their professional growth over time. The seamless connection between

these phases ensures that Pathfinder provides not only initial career guidance but also ongoing support for users' professional development journeys.

3.6 RESPONDENTS OF THE STUDY

The study engaged participants directly involved in the academic and career development ecosystem to ensure the Pathfinder system's recommendations are technically robust, relevant, and user-friendly. The primary respondents consisted of at least 20 - 50 Filipino college students recruited from any university. These students represented a diverse range of academic disciplines and year levels to capture a wide variety of profiles and career planning needs.

Their participation was crucial for evaluating the user experience (UX), interface usability, and the perceived relevance and accuracy of the career recommendations. Complementing this group, at least five (5) licensed guidance counselors or psychologist and career advisors were involved in the study. Their expertise provided professional validation of the assessment's psychological basis (MBTI) and the overall utility of the system as a counseling aid. Furthermore, at least five (5) industry professionals from various sectors in the Philippine job market were consulted to assess the alignment of the system's recommendations with real-world industry demands and trends.

Data collection methods were multi-faceted. Students participated in user testing sessions where they completed the assessment and interacted with the recommendation dashboard; their interactions were anonymized and logged. Structured surveys and focus group discussions (FGDs) were conducted with all respondent groups to gather qualitative feedback on the system's explainability, fairness, and practical value. Ethical considerations were rigorously upheld. All participants provided informed consent after a detailed explanation of the study's objectives, data usage, and privacy protocols. All collected data was anonymized, and participation was voluntary, with the option to withdraw at any time without penalty.